

Training Configuration and Summary:

Dataset used: Training: rssi_odom_dataset_11apr25_isolado.parquet | Experiment: rssi_odom_dataset_12apr25_isolado.parquet

model_type: GRU
group_sec: 40
timesteps_orig: 60
batch_size: 256
min_units: 16
max_units: 1024
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
group_msec: 40
samples_per_sec: 25
forecast_horizon_sec: 4
forecast_steps: 100
timesteps: 60

Dataset Parameters:

Training: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60
Experiment: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60

Tuning:

Trials executed: 10/10

Training Performance:

Loss: 0.0035 | MAE: 0.0466

Validation Performance:

Loss: 0.0119 | MAE: 0.0862

GRU Multi-Step Training Report

Report Generated: 2025-09-14 01:00:05

Training Start: 2025-09-13 20-09-32

Training End: 2025-09-14 00-39-13

Training Duration: 269.68 minutes

Predictions Start: 2025-09-14 00-39-13

Predictions End: 2025-09-14 00-49-46

Predictions Duration: 10.55 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 2.8415

MSE: 12.7093

RMSE: 3.5650

R2 Score: 0.6164

Multi-Step Forecast:

Forecast Steps: 100

MAE: 0.0482

MSE: 0.0036

RMSE: 0.0596

R2 Score: 0.9167

Dataset size before down-sample: 31,097 rows
Dataset size after down-sample: 9,344 rows

First 5 timestamps BEFORE:

1744464597710
1744464597720
1744464597744
1744464597757
1744464597783

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

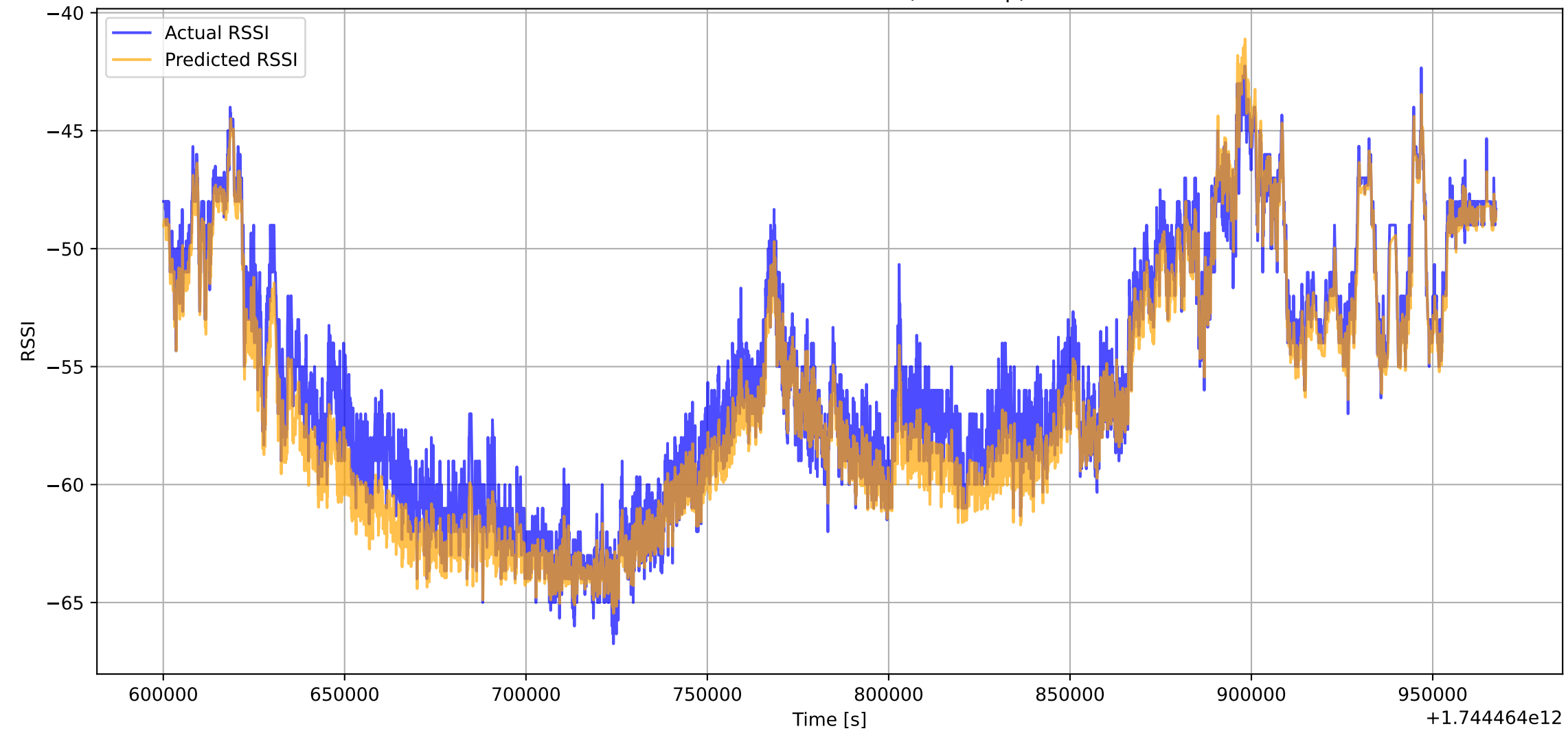
1744464597710
1744464597720
1744464597783
1744464597812
1744464597879

...

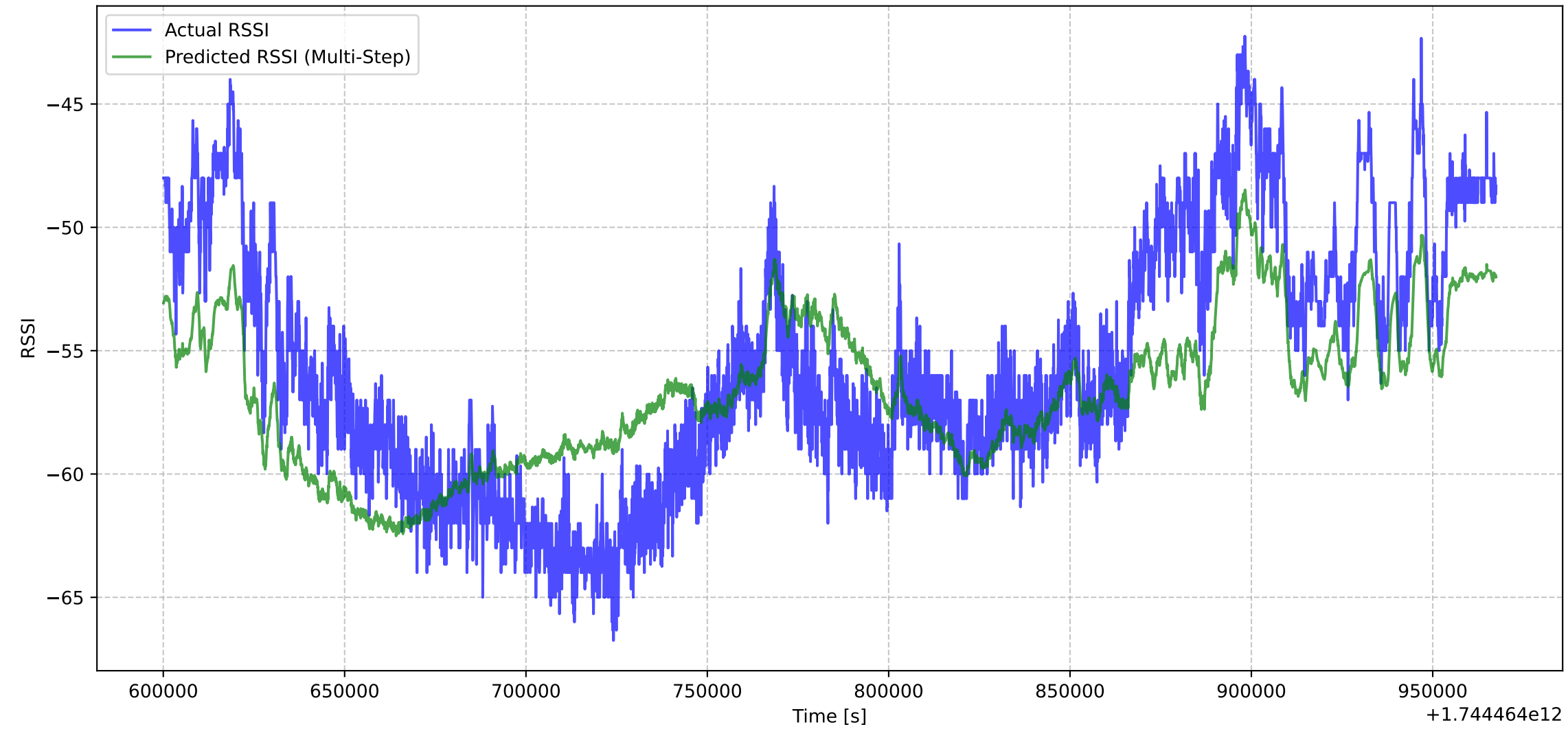
Last 5 timestamps AFTER:

1744464971245
1744464971281
1744464971329
1744464971366
1744464971402

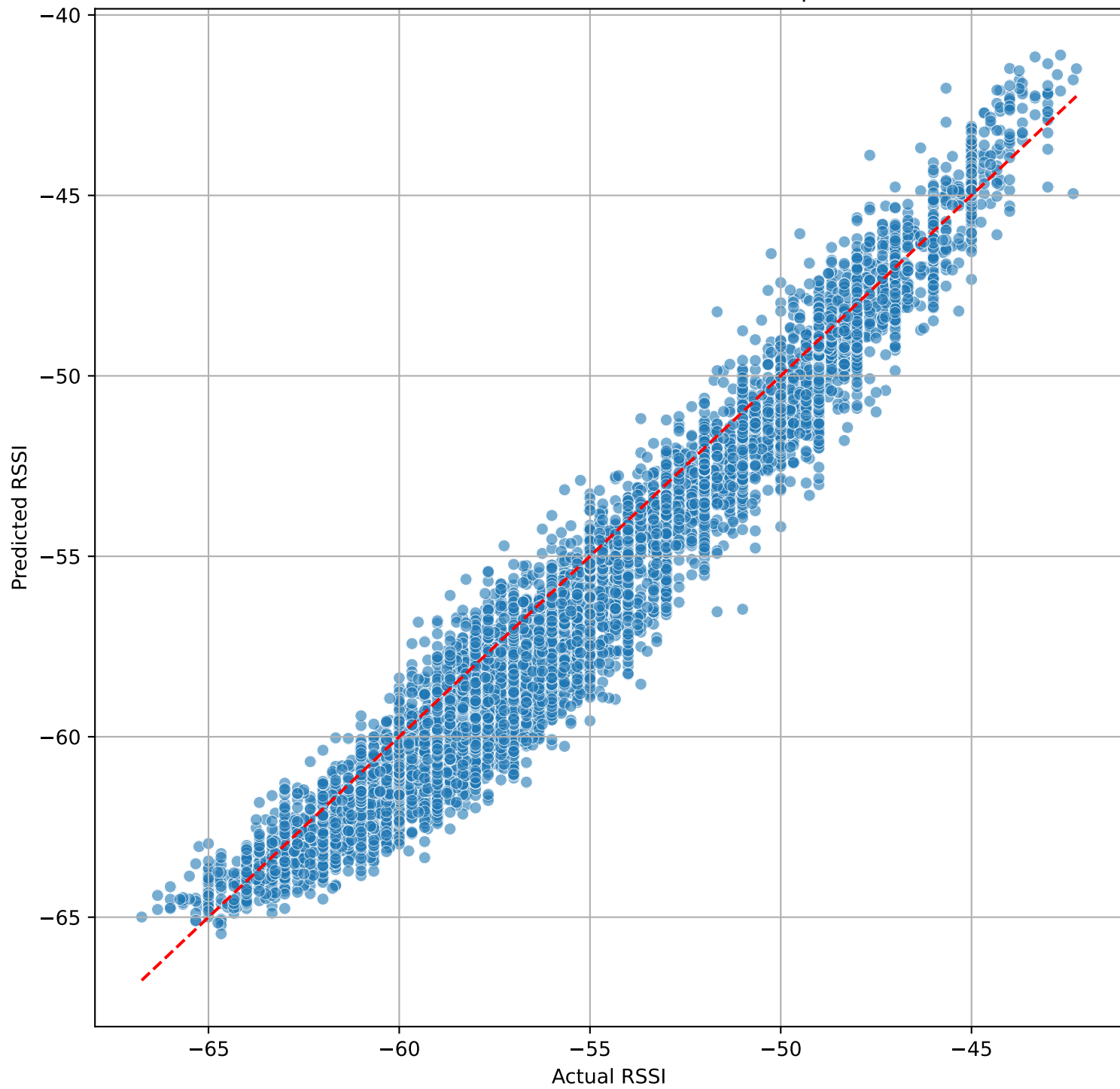
Actual vs Predicted RSSI (First Step)



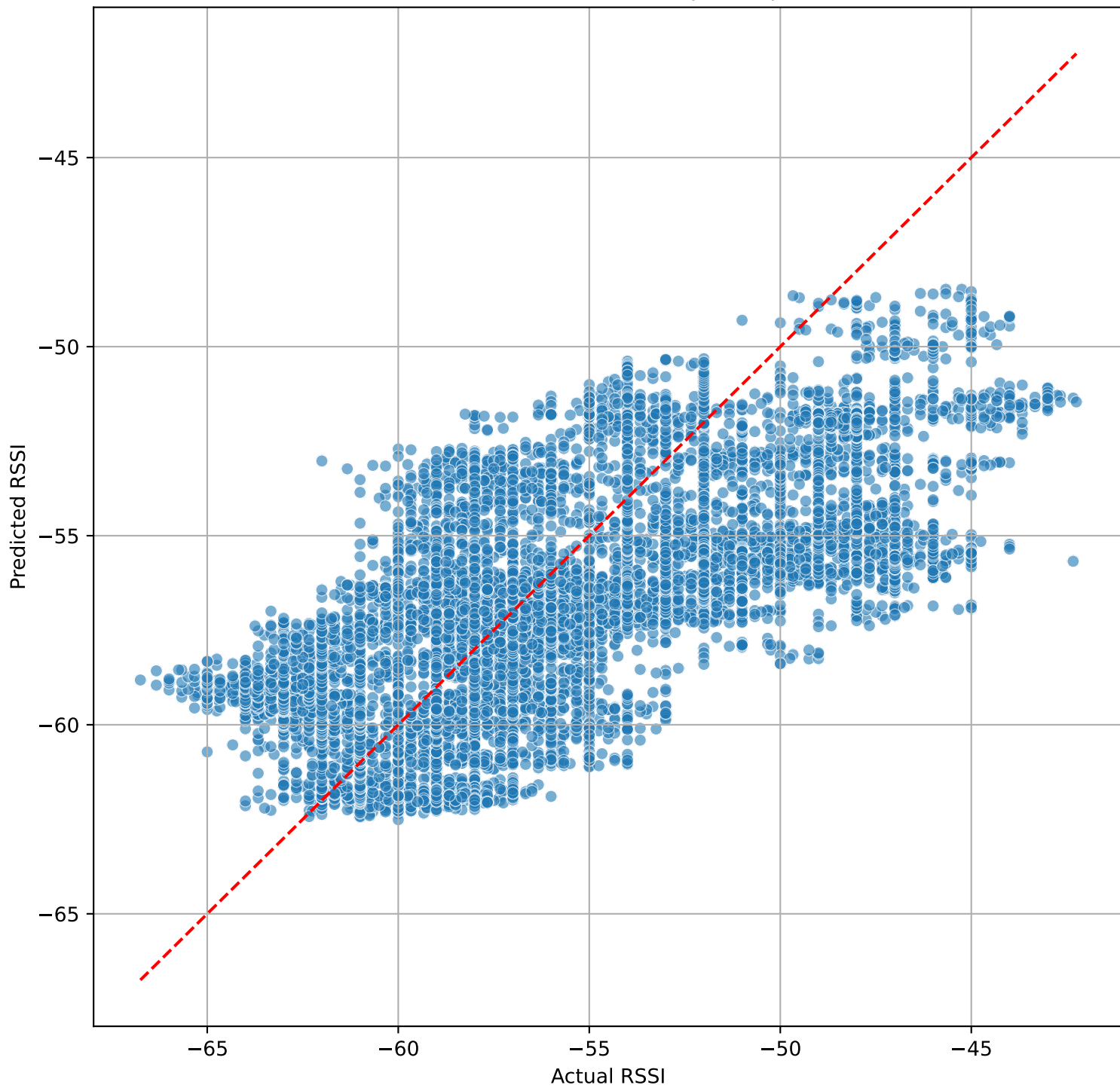
Actual vs Predicted RSSI (Multi-Step (step 100 $\approx$ 4.0 s))



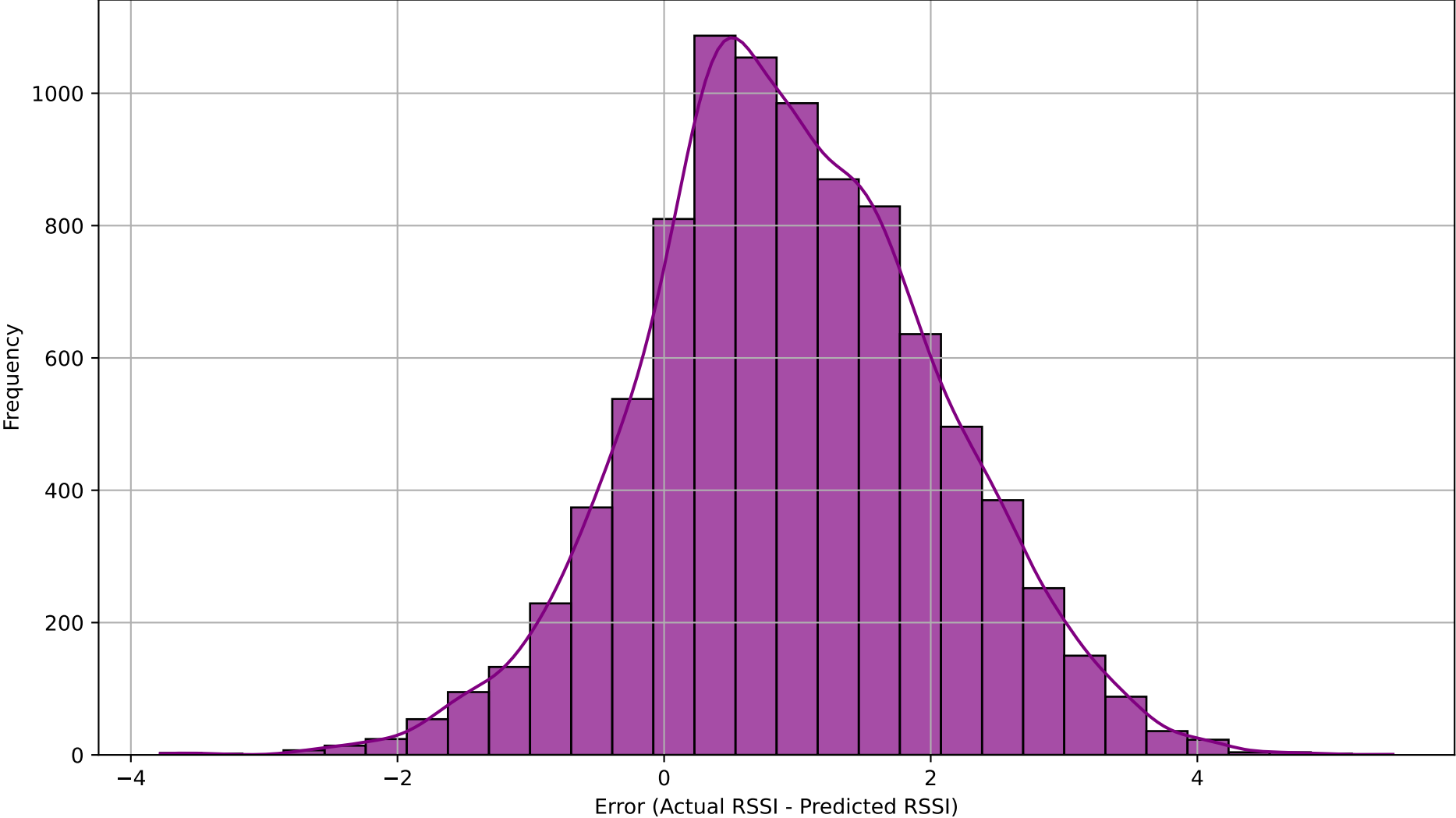
Actual vs Predicted RSSI (First Step)



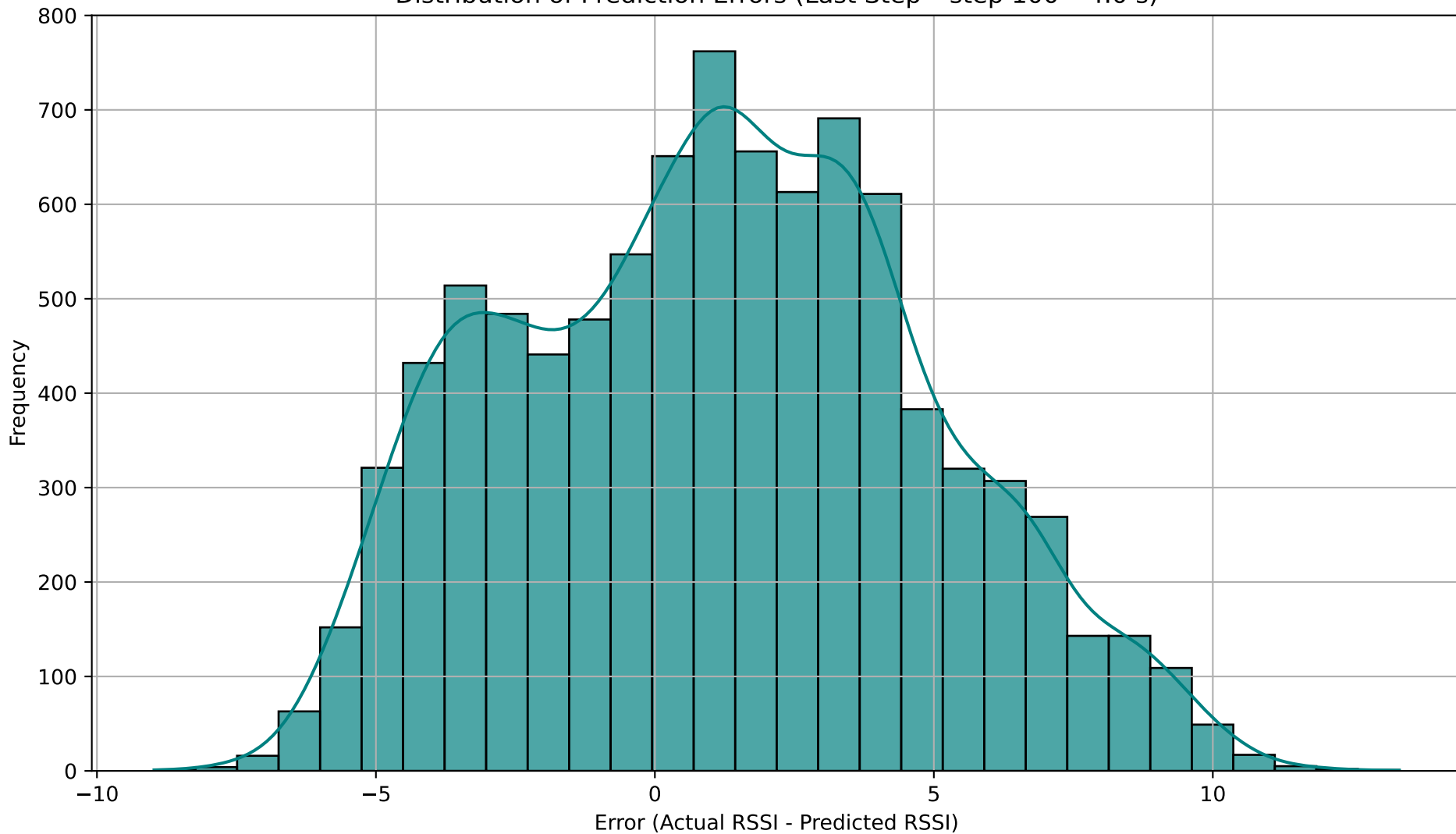
Actual vs Predicted RSSI (Last Step - step 100 $\approx$ 4.0 s)



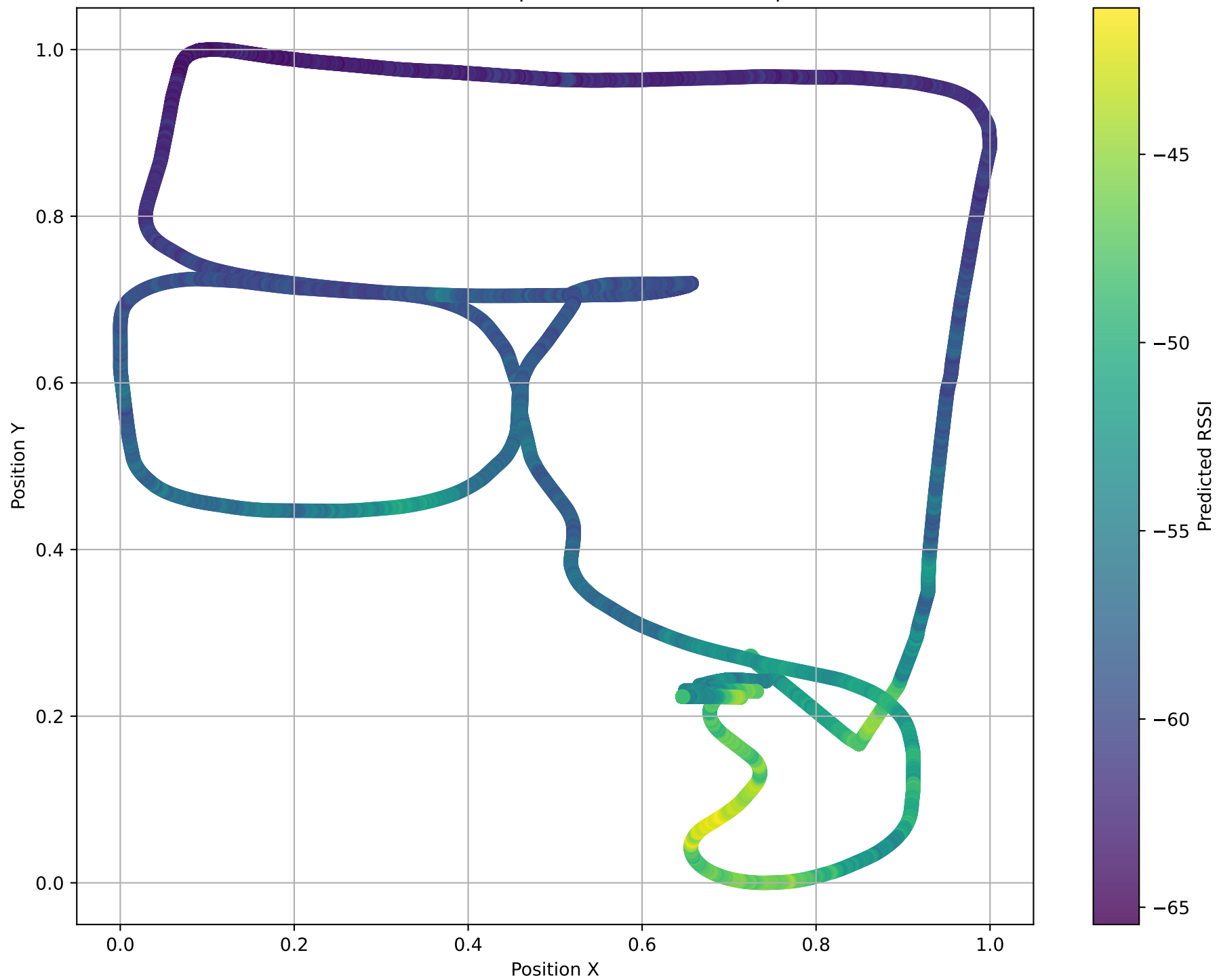
Distribution of Prediction Errors (First Step)



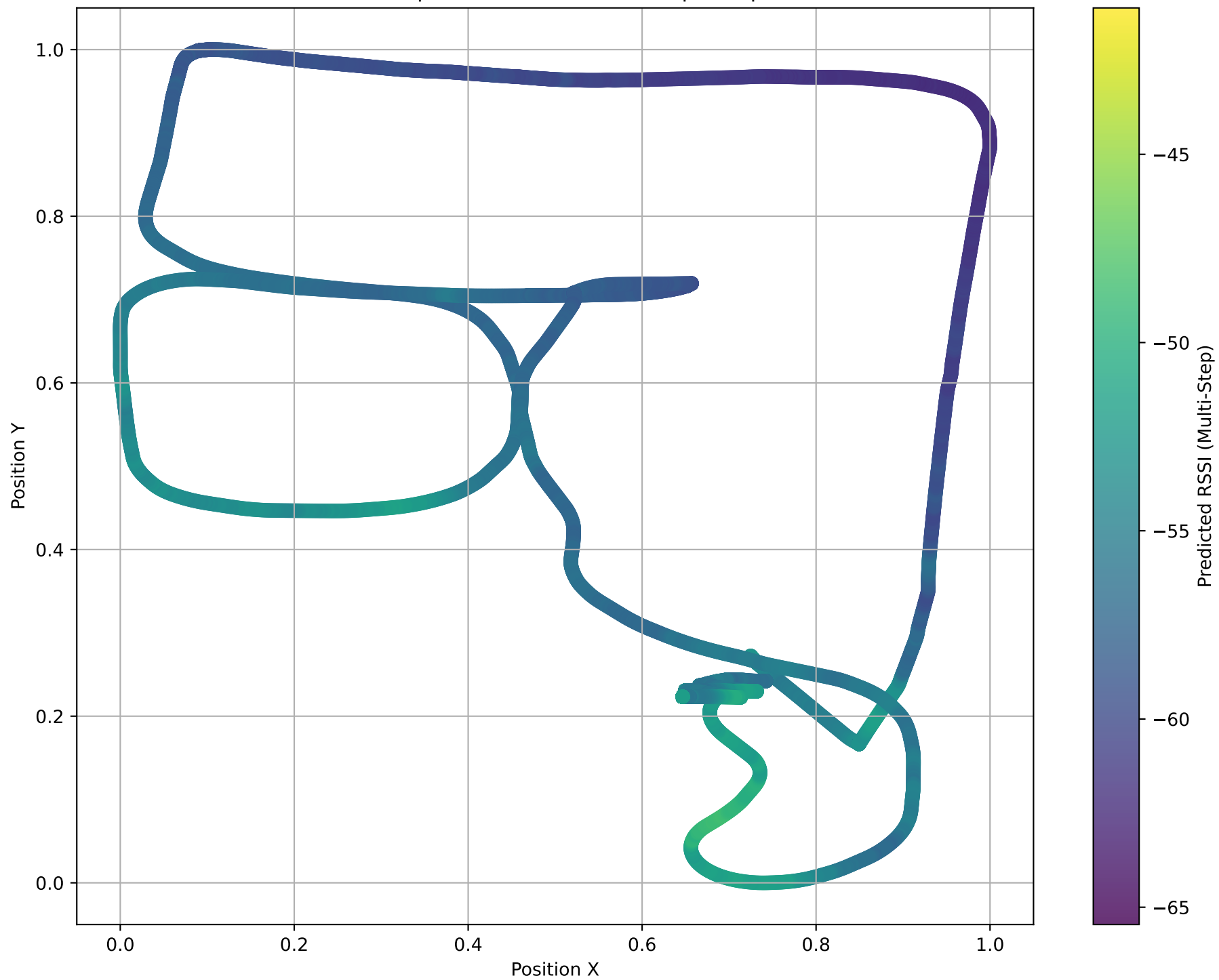
Distribution of Prediction Errors (Last Step - step 100 $\approx$ 4.0 s)



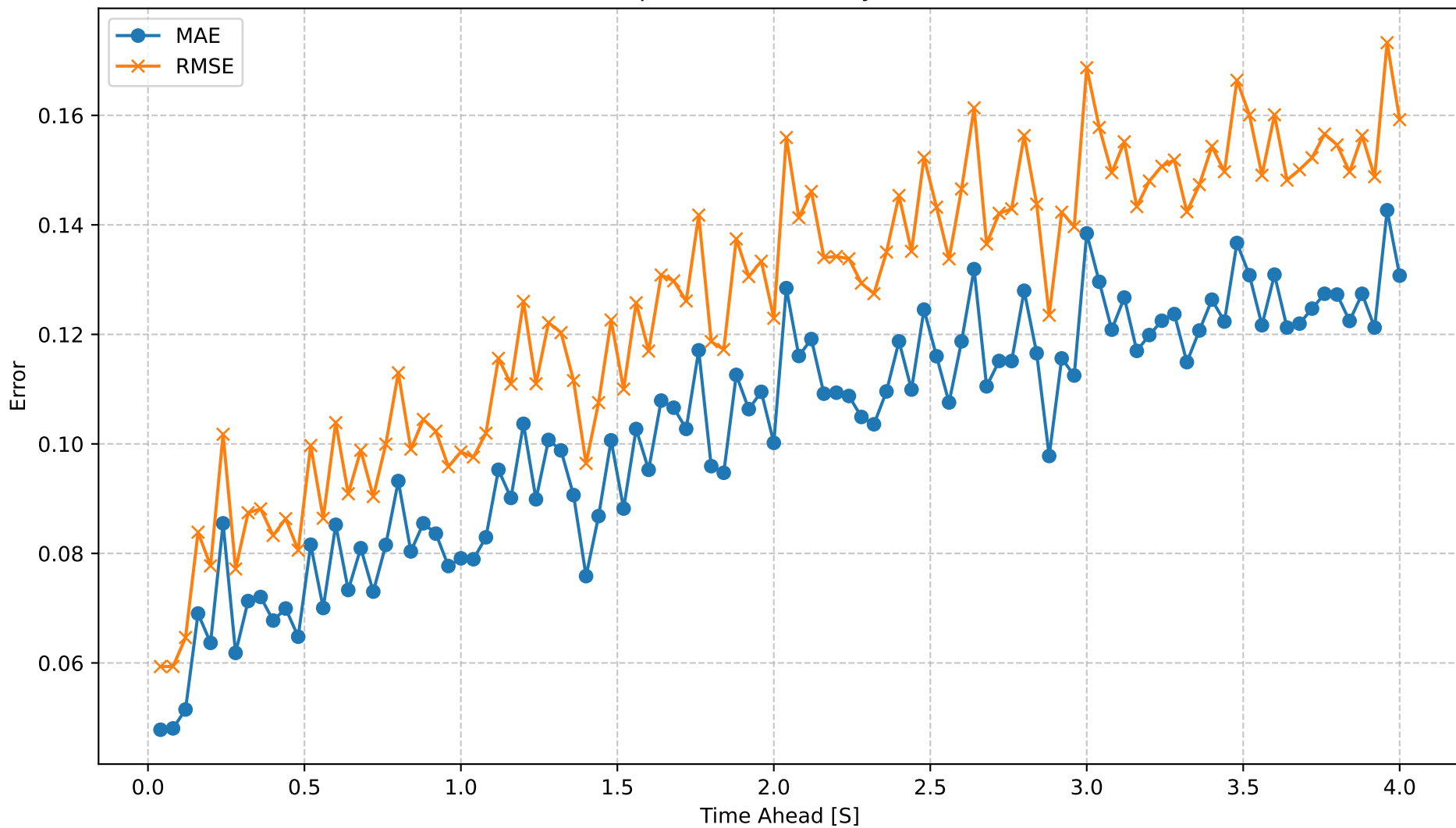
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 100 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: Training: rssi_odom_dataset_11apr25_isolado.parquet |
Experiment: rssi_odom_dataset_12apr25_isolado.parquet
Modelo: GRU
Forecast steps: 100

MAE / RMSE (dB)

- 1º passo : 0.05 / 0.06
- Mediana : 0.11 / 0.13
- 100º passo : 0.13 / 0.16

Crescimento do erro: 173.5% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 99 (0.14), 75 (0.14), 87 (0.14)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.